

Exam FAM Formula Sheet

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Every formula below is derived and explained in full in the Exam FAM Study Manual. This page exists purely for fast lookup and final-week memorization, condensed to fit a short printout. Nothing here is a substitute for understanding where each formula comes from.

1. Short-Term Insurance & Reinsurance Coverages (5–10%)

PAYMENT R.V. (DEDUCTIBLE D , LIMIT U , COINSURANCE α)

$$Y^L = \alpha \cdot [\min(X, u) - \min(X, d)]$$

LOSS ELIMINATION RATIO

$$\text{LER} = E[X \wedge d] / E[X]$$

PER-PAYMENT VS. PER-LOSS VARIABLE

$$Y^P = Y^L \mid X > d$$

EXCESS-OF-LOSS REINSURER'S SHARE

 $\min(\max(X - r, 0), m)$ (layer from retention r , width m)

QUOTA SHARE

Insurer retains $\alpha \cdot X$; Reinsurer pays $(1 - \alpha) \cdot X$

2. Severity, Frequency & Aggregate Models (12.5–17.5%)

Severity Distributions

COMMON SEVERITY DISTRIBUTIONS — MEAN AND VARIANCE

Distribution	$E[X]$	$\text{Var}(X)$	Notes
Exponential(θ)	θ	θ^2	special case: Weibull($\tau=1$), Gamma($\alpha=1$)
Gamma(α, θ)	$\alpha\theta$	$\alpha\theta^2$	sum of α iid Exponential(θ) when $\alpha \in \mathbb{Z}^+$
Pareto(α, θ)	$\theta/(\alpha-1)$, $\alpha > 1$	$\alpha\theta^2 / [(\alpha-1)^2(\alpha-2)]$, $\alpha > 2$	$E[X^k]$ exists only for $k < \alpha$; heavy-tailed, no MGF
Weibull(τ, θ)	$\theta\Gamma(1+1/\tau)$	$\theta^2 [\Gamma(1+2/\tau) - \Gamma(1+1/\tau)^2]$	$E[X^k] = \theta^k \Gamma(1+k/\tau)$; $\tau=1$ reduces to Exponential
Lognormal(μ, σ^2)	$e^{(\mu+\sigma^2/2)}$	$e^{(2\mu+\sigma^2)}(e^{\sigma^2} - 1)$	$\ln X \sim \text{Normal}(\mu, \sigma^2)$; no MGF, heavy-tailed

PARETO PDF / SURVIVAL

$$f(x) = \alpha\theta^\alpha / (x+\theta)^{(\alpha+1)}, S(x) = [\theta/(x+\theta)]^\alpha$$

EXISTENCE OF MOMENTS (PARETO)

$$E[X^k] \text{ finite} \Leftrightarrow k < \alpha$$

Frequency: The (a,b,0) Class

(A,B,0) RECURSION

$$p_k/p_{k-1} = a + b/k, k=1,2,3,\dots$$

(A,B,0) MEMBERS — PARAMETERS, MEAN, VARIANCE

Distribution	a	b	E[N]	Var(N)
Poisson(λ)	$\emptyset$	λ	λ	λ
Binomial(m,q)	$-q/(1-q)$	$(m+1)q/(1-q)$	mq	$mq(1-q)$
Negative Binomial(r, β)	$\beta/(1+\beta)$	$(r-1)\beta/(1+\beta)$	$r\beta$	$r\beta(1+\beta)$

Aggregate Loss Models

COLLECTIVE RISK MODEL

$$S = X_1 + \dots + X_N, E[S] = E[N]E[X], \\ \text{Var}(S) = E[N]\text{Var}(X) + \text{Var}(N)E[X]^2$$

INDIVIDUAL RISK MODEL

$$S = Y_1 + \dots + Y_n, Y_i \text{ independent (not identical)}$$

NORMAL APPROXIMATION

$$P(S \leq s) \approx \Phi[(s - E[S]) / \sqrt{\text{Var}(S)}]$$

LOGNORMAL APPROXIMATION (MOMENT MATCHING)

$$\sigma^2 = \ln(1 + \text{Var}(S)/E[S]^2), \mu = \ln(E[S]) - \sigma^2/2$$

EXPECTED STOP-LOSS PAYMENT

$$E[(S-d)^+] = E[S] - E[S \wedge d]$$

VALUE AT RISK

$$\text{VaR}_p(S) = \inf\{s : F_S(s) \geq p\}$$

TAIL VALUE AT RISK

$$\text{TVaR}_p(S) = \text{VaR}_p(S) + E[(S - \text{VaR}_p(S))^+] / (1-p)$$

3. Parametric Estimation (2.5-7.5%)

LIKELIHOOD — COMPLETE DATA

$$L(\theta) = \prod f(x_i; \theta)$$

LIKELIHOOD — GROUPED DATA

$$L(\theta) = \prod [F(c_{j+1}; \theta) - F(c_j; \theta)]^{(n_j)}$$

LIKELIHOOD — LEFT TRUNCATION AT D

$$\text{Contribution} = f(x; \theta) / S(d; \theta), x > d$$

LIKELIHOOD — RIGHT CENSORING AT U

$$\text{Contribution} = S(u; \theta)$$

4. Introduction to Credibility (2.5-5%)

CREDIBILITY PREMIUM

$$P = Z \cdot (\text{observed}) + (1-Z) \cdot (\text{prior/manual})$$

STANDARD FOR FULL CREDIBILITY (CLAIM COUNTS)

$$\lambda_0 = (z_{((1+p)/2)} / k)^2$$

PARTIAL CREDIBILITY

$$Z = \sqrt{(n/n_0)}, \text{ capped at } 1$$

5. Pricing & Reserving for Short-Term Coverages (10-15%)

EXPECTED LOSS RATIO RESERVE

$$\text{Ultimate Losses} = \text{ELR} \times \text{Earned Premium}$$

CHAIN-LADDER AGE-TO-AGE FACTOR

$$f_k = \Sigma(\text{losses at age } k+1) / \Sigma(\text{losses at age } k), \text{ volume-weighted}$$

BORNHUETTER-FERGUSON

$$\text{Ultimate} = \text{Actual-to-date} + (1 - 1/\text{CDF}) \times (\text{Expected Losses})$$

PERMISSIBLE LOSS RATIO

$$\text{PLR} = 1 - \text{Variable Expense Ratio} - \text{Target Profit \%}$$

LOSS RATIO METHOD

$$\text{Indicated Rate Change} = (\text{Experience Loss Ratio} / \text{PLR}) - 1$$

PURE PREMIUM METHOD

$$\text{Indicated Rate} = \text{Projected Pure Premium} / \text{PLR}$$

6. Option Pricing Fundamentals (2.5-7.5%)

CALL / PUT PAYOFF

$$\text{Call: } \max(S_T - K, 0) \quad \text{Put: } \max(K - S_T, 0)$$

BINOMIAL RISK-NEUTRAL PROBABILITY

$$p^* = (e^{rh} - d) / (u - d)$$

BINOMIAL OPTION PRICE

$$V_0 = e^{-rh} [p^* \cdot V_u + (1 - p^*) \cdot V_d]$$

BLACK-SCHOLES CALL PRICE

$$C = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

D1, D2

$$d_1 = [\ln(S_0/K) + (r + \sigma^2/2)T] / (\sigma\sqrt{T}), d_2 = d_1 - \sigma\sqrt{T}$$

DELTA

$$\Delta_{\text{call}} = N(d_1), \Delta_{\text{put}} = N(d_1) - 1$$

PUT-CALL PARITY

$$C - P = S_0 - Ke^{-rT}$$

7. Mortality Models (10-15%)

SURVIVAL PROBABILITY

$${}_t p_x = S(x+t)/S(x) = l_{x+t}/l_x$$

FORCE OF MORTALITY

$$\mu(x) = -S'(x)/S(x), S(x) = \exp[-\int_0^x \mu(s) ds]$$

DE MOIVRE

$$\mu(x) = 1/(\omega - x), {}_t p_x = (\omega - x - t)/(\omega - x)$$

CONSTANT FORCE (EXPONENTIAL)

$$\mu(x) = \mu, {}_t p_x = e^{-\mu t}$$

GOMPERTZ / MAKEHAM

$$\mu(x) = Bc^x \text{ (Gompertz)}, \mu(x) = A + Bc^x \text{ (Makeham)}$$

COMPLETE / CURTATE EXPECTATION

$$\dot{e}_x = \int_0^\infty {}_t p_x dt, e_x = \sum_{k \geq 1} {}_k p_x, \dot{e}_x \approx e_x + 0.5 \text{ (UDD)}$$

UDD FRACTIONAL AGE

$${}_sq_x = s \cdot q_x, 0 \leq s \leq 1$$

8. Present Value Random Variables (12.5–20%)

WHOLE LIFE INSURANCE (DISCRETE)

$$Z = v^{K_x+1}, A_x = E[Z] = \sum_{k \geq 0} v^{k+1} \cdot {}_kp_x \cdot q_{x+k}$$

WHOLE LIFE INSURANCE (CONTINUOUS)

$$Z = v^{T_x}, \bar{A}_x = \int_0^\infty v^t \cdot {}_tp_x \cdot \mu(x+t) dt$$

WHOLE LIFE ANNUITY-DUE

$$\ddot{a}_x = \sum_{k \geq 0} v^k \cdot {}_kp_x$$

INSURANCE-ANNUITY IDENTITY

$$A_x = 1 - d \cdot \ddot{a}_x \text{ (discrete); } \bar{A}_x = 1 - \delta \cdot \bar{a}_x \text{ (continuous)}$$

VARIANCE — INSURANCE

$$\text{Var}(Z) = {}^2A_x - (A_x)^2 \text{ (} {}^2A_x \text{ uses double the force of interest, i.e. } v^2 \text{)}$$

VARIANCE — ANNUITY

$$\text{Var}(Y) = [{}^2A_x - (A_x)^2] / d^2$$

9. Premium & Policy Value Calculation (15–22.5%)

FUTURE LOSS R.V. (WHOLE LIFE)

$$L_0 = v^{K_x+1} - P \cdot \ddot{a}_{x+1}$$

EQUIVALENCE PRINCIPLE PREMIUM

$$E[L_0] = 0 \Rightarrow P = A_x / \ddot{a}_x = (1 / \ddot{a}_x) - d \text{ (whole life)}$$

PROSPECTIVE POLICY VALUE

$${}_tV = \text{EPV}(\text{future benefits}) - \text{EPV}(\text{future net premiums})$$

RECURSIVE RESERVE RELATIONSHIP

$$({}_tV + P)(1+i) = q_{x+t} \cdot (\text{benefit}) + p_{x+t} \cdot {}_{t+1}V$$

EXTRA RISK — CONSTANT ADDITION

$$\mu^*(x) = \mu(x) + c$$

EXTRA RISK — CONSTANT MULTIPLE

$$q^*(x) = k \cdot q(x)$$

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